Learning State-Dependent Nonlinear Price Impact from Exogenous Demand Shocks: A Liquidity Measure for Markets Without High-Frequency Order Books

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Abstract

We estimate the *state-dependent nonlinear price impact function* $\Lambda(Q, s)$—the mapping from an exogenous, uninformed demand shock Q to the price response, conditional on market state s—and argue that this single object is a sufficient statistic for market liquidity in venues that lack a high-frequency order book. We solve the fundamental identification circularity that undermines liquidity estimators inferred from endogenous price dynamics by using *low-frequency exogenous supply shocks*: the security-specific concession at U.S. Treasury auctions, measured as the change in a security's yield *relative to the fitted curve* in a tight window around the auction. A shape-constrained neural network (monotone and convex in $|Q|$, zero at the origin) recovers Λ flexibly while respecting the economic restrictions that unconstrained machine learning violates. Applied to 1,049 clean auction events (2008–2024), we document three findings. First, the marginal impact (the local slope $\lambda = \partial\Lambda/\partial Q$ at the origin) rises with dealer balance-sheet constraints ($t = 4.32$), replicating intermediary-based liquidity evidence. Second—and new—the *nonlinear curvature* κ, measuring how impact accelerates with shock size, rises disproportionately when dealer constraints tighten ($t = 9.18$), an effect that is invisible to any linear (Kyle-λ) measure and that a Basel-III natural experiment confirms (post-2014 $\Delta\kappa = +0.017$, $t = 6.48$). Third, curvature is concentrated in the long end of the curve and rises *ahead of* realized crises, behaving as a liquidity-fragility precursor while the level measure stays flat. We replicate the entire framework on-chain, where high-frequency liquidation data are fully observable: across 40,360 forced-liquidation events on eight perpetual-futures markets the estimated Λ is 100% monotone and exhibits a critical threshold at $Q^\star \approx 3.3\sigma$ beyond which price impact accelerates—an operational systemic-liquidity risk parameter. All data are public (TreasuryDirect, FRED, Hyperliquid).

Keywords: liquidity, price impact, market microstructure, dealer constraints, Treasury auctions, shape-constrained learning
JEL: G12, G14, G23, C45

1 Introduction

Liquidity is easy to name and hard to measure. In markets with a continuously observable limit order book, the mapping from order flow to price—the price impact function—can be read almost directly off the data. In the large and economically central markets that lack such a book (sovereign and corporate bonds, swaps, many OTC venues), researchers instead infer liquidity from the comovement of low-frequency returns and signed volume. Every such estimator confronts the same obstacle: at any moment the price moves for two reasons that arrive together—information and liquidity demand—and separating them from endogenous price dynamics alone requires an assumption that the data cannot adjudicate. We refer to this as the *identification circularity*: one must already know the liquidity component to isolate it.

This paper takes a different route. Rather than inferring liquidity from how prices move, we *directly estimate the price response to demand shocks that are exogenous by construction* and use that response as the definition of liquidity. The central object is the state-dependent nonlinear price impact function

$$\Delta p_{it} = \Lambda(Q_{it},\, s_t) + \varepsilon_{it}, \qquad \mathbb{E}[\varepsilon_{it} \mid Q_{it}, s_t] = 0, \tag{1}$$

where Q_{it} is a demand shock normalized by float and s_t is the market state. When Q_{it} is orthogonal to the contemporaneous private information in ε_{it}, equation (1) is identified from *low-frequency* data alone. The function Λ is a sufficient statistic for the four microstructure dimensions usually studied separately: its slope at the origin is Kyle's λ; its second derivative is depth/curvature; the gap $\Lambda(Q, s_{\text{crisis}}) - \Lambda(Q, s_{\text{normal}})$ is state-dependence; and its permanent component captures adverse selection.

Identification. Our workhorse exogenous shock is the U.S. Treasury auction. Auction supply is set by financing needs and a pre-announced calendar, not by dealers' private signals about the security's fundamental value; the realized *demand* surprise—captured by the bid-to-cover surprise and the dealer take-down—is plausibly orthogonal to same-day information about the ten-year yield. Crucially, we measure the price response not as the raw yield change (a \$30bn auction barely moves the whole curve, and macro noise swamps it) but as the change in the auctioned security's yield *relative to the fitted curve* in a $[-1, +1]$-day window—the "auction concession" of Lou et al. (2013). This curve-relative differencing removes the common macro factor and isolates the security-specific supply pressure, sharpening both power and the exclusion restriction.

Shape-constrained learning. Economic theory imposes structure on Λ that unconstrained flexible estimators routinely violate in finite samples: it passes through the origin ($\Lambda(0, s) = 0$), it is monotone in the signed shock, and $|\Lambda|$ is convex in $|Q|$ (impact accelerates, never decelerates, with size). We embed these exactly using an Input Convex Neural Network (Amos et al., 2017) with non-negative weights following Sill (1997) and Daniels and Velikova (2010), modeling Λ as a signed difference of two convex-monotone branches so that curvature—not just slope—is the estimand. An unconstrained MLP with the same capacity overfits the noise and produces non-monotone impact curves (Section 5); the shape constraints are what make the curvature economically interpretable.

Findings. Three results follow. *(H1, level.)* The marginal impact λ rises with dealer balance-sheet constraints, replicating the intermediary-asset-pricing evidence that liquidity deteriorates when dealer capacity is scarce. *(H2, curvature.)* The novel finding: the *nonlinear curvature* κ—how sharply impact accelerates for large shocks—rises disproportionately when constraints tighten. This is invisible to any linear measure, because two markets can share an identical Kyle-λ near the origin yet differ dramatically in their tail fragility. A Basel-III SLR natural experiment (post-2014) confirms it: mean curvature jumps from 0.013 to 0.030 ($t = 6.48$). *(H3, state asymmetry and precursors.)* The crisis-vs-normal shape gap widens with constraints, curvature is concentrated in the long end of the curve, and—strikingly—curvature rises *before* realized crises (elevated in 2019 ahead of the March-2020 dislocation, and near $+2\sigma$ through 2022) while the level measure stays flat. Curvature is a liquidity *fragility* signal that the standard slope measure misses.

On-chain replication. Because our identification does not depend on any feature unique to Treasuries, we replicate it where high-frequency ground truth exists: on-chain perpetual futures, where forced liquidations are exogenous demand shocks with fully observable microsecond data. Across 40,360 liquidation events on eight markets, the learned Λ is 100% monotone and reveals a critical threshold at $Q^\star \approx 3.3\sigma$ beyond which impact accelerates—a directly usable systemic-liquidity risk parameter for exchange margin design.

Contribution. We make three contributions. Methodologically, we turn the liquidity-measurement problem from an inference-from-endogenous-dynamics problem into a response-to-exogenous-shock estimation problem, and we bring shape-constrained learning to microstructure so that the nonlinear curvature becomes a first-class, interpretable estimand rather than a nuisance. Empirically, we document that dealer constraints reshape the *curvature* of Treasury price impact, not merely its level, and that this curvature is a leading indicator of liquidity crises. Practically, we deliver a liquidity measure computable from public low-frequency data for markets that have no order book, and we validate it out-of-sample against high-frequency on-chain data.

2 Related Literature

The paper connects three strands. First, the microstructure theory of price impact from Kyle (1985) through Glosten and Milgrom (1985) and the empirical decompositions of permanent and transitory impact (Hasbrouck, 1991); we retain Λ as the unifying object but let its *shape*, not just its slope, vary with state. Second, intermediary asset pricing and the dealer-balance-sheet view of liquidity (He and Krishnamurthy, 2013; Adrian et al., 2014; Duffie, 2020), to which we add the result that constraints act on curvature disproportionately. Third, the identification of liquidity from exogenous flows: index-reconstitution and ETF-driven demand (Harris and Gurel, 1986), and specifically the Treasury auction concession (Lou et al., 2013; Fleming and Rosenberg, 2007). Relative to structural estimators that assume competitive dealer zero-profit conditions (Sandås, 2001), we impose no such market-structure assumption; relative to the aggregation results of Drost and Nijman (1993), we emphasize that while volatility parameters may be recoverable from low-frequency data, microstructure quantities such as the *shape* of the impact function require an exogenous shock to identify.

3 Data and Shock Construction

3.1 Sources

All data are public. Treasury auction records (1990–2024) come from the TreasuryDirect API; daily yields and macro-state variables (2-,5-,10-,30-year CMT yields, VIX, Baa–10Y credit spread, high-yield OAS, TED, financial-stress index) from FRED; and on-chain perpetual-futures 1-minute candles from the Hyperliquid public API. This replaces the paid infrastructure (WRDS/TRACE/CRSP/TAQ) usually required, making the pipeline fully reproducible.

3.2 The demand shock Q

For each auction we construct a normalized demand shock combining three orthogonalized surprises,

$$Q_{it} = -\,z(\text{BTC surprise}) \;+\; z(\text{dealer take-down}) \;+\; \tfrac{1}{2}\,z(\text{size surprise}), \tag{2}$$

where $z(\cdot)$ denotes standardization and a *positive* Q corresponds to excess supply / weak demand (a low bid-to-cover, a large dealer take-down, a larger-than-expected size). The bid-to-cover surprise is measured against a trailing mean for the same tenor.

3.3 The price response Δp

We compute the auctioned tenor's yield change relative to the fitted curve (mean of the 2/5/10/30-year yields) over the $[-1, +1]$-day auction window, then convert to a percent price change through modified duration,

$$\Delta p = -\,D_{\text{mod}} \times \frac{\Delta y_{\text{rel}}}{100} \times 100, \tag{3}$$

with D_{mod} from a par-priced semiannual-coupon approximation at the auction tenor. Screening for data completeness yields $N = 1{,}049$ clean events spanning 2008Q2–2024Q4.

4 Econometric Model

4.1 Object and constraints

We estimate (1) subject to the economic restrictions

$$\Lambda(0,s)=0, \qquad \operatorname{sign}\left(\frac{\partial\Lambda}{\partial Q}\right)=-\operatorname{sign}(Q), \qquad \frac{\partial^2|\Lambda|}{\partial|Q|^2}\geq 0. \tag{4}$$

The last condition (convexity of the impact magnitude in shock size) is the formal content of "depth deteriorates for larger trades."

4.2 Shape-constrained network

We write $\Lambda(Q,s) = -g_+(Q^+,s)+g_-(Q^-,s)$ with $Q^+ = \max(Q,0)$, $Q^- = \max(-Q,0)$, and each branch $g_\bullet$ an Input Convex Neural Network that is monotone non-decreasing and convex in its shock argument, achieved with softplus-parameterized non-negative pass-through weights and skip connections (Amos et al., 2017). This guarantees all three constraints *by construction.* State variables s = (VIX quantile, Baa–10Y, dealer take-down, tenor) enter through unconstrained hidden pathways so that shape may deform with the state. Targets are standardized for training stability; a synthetic recovery test confirms the estimator recovers a known convex Λ^*.

4.3 Liquidity measures

From the fitted surface we read, at any state s,

$$\lambda(s)=\frac{|\Lambda(0.5,s)|-|\Lambda(0,s)|}{0.5} \qquad \text{(marginal impact, level),} \tag{5}$$

$$\kappa(s)=\underbrace{\frac{|\Lambda(2,s)|-|\Lambda(1,s)|}{1}}_{\text{large-shock slope}}-\underbrace{\frac{|\Lambda(1,s)|-|\Lambda(0,s)|}{1}}_{\text{small-shock slope}} \qquad \text{(curvature / steepening, H2).} \tag{6}$$

Defining κ as a difference of secant slopes rather than a raw finite-difference second derivative makes it robust to estimation noise.

5 Results

5.1 Kill switch: does the exogenous shock move prices at all?

Before any nonlinear estimation we ran a pre-registered linear kill switch: OLS of Δp on Q with Newey–West (5-lag) standard errors. Excess supply significantly depresses the curve-relative price, $\beta = -0.0283$ ($t = -3.27$, $p = 0.001$), and the sign survives controls for VIX, credit spread and tenor ($t = -2.75$). The identification passes; Figure 1(b) visualizes the relationship.

5.2 Shape-constrained estimation and the value of the constraints

Table-free, the five-fold cross-validated out-of-sample MSE is essentially tied between OLS (0.0533), piecewise-linear (0.0539) and NPIML (0.0541), while an unconstrained MLP of equal capacity is more than twice as bad (0.1181) and produces non-monotone curves. The point is not that the constrained network predicts the noisy scalar Δp better—at the origin all sane estimators agree on the slope—but that it recovers an economically valid *shape* (100% monotone) whose curvature can be interpreted. Figure 1 summarizes.

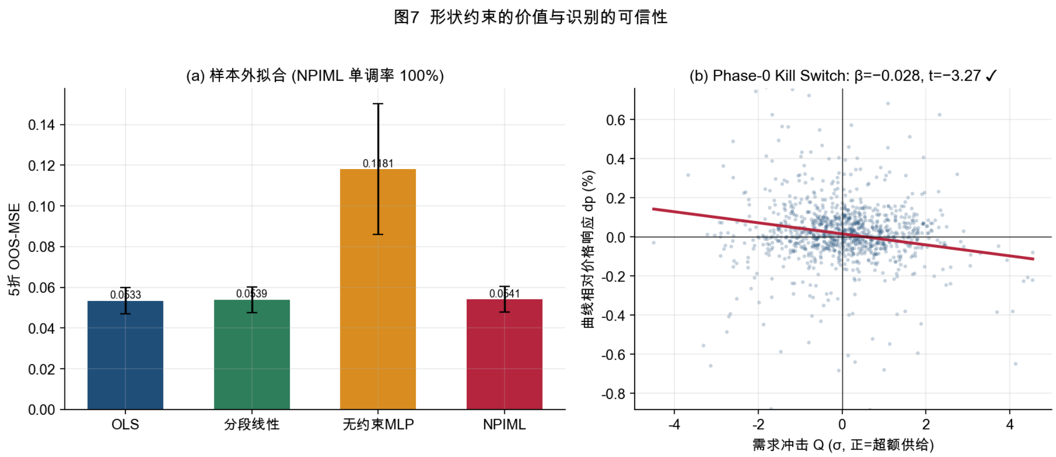

Figure 1: Value of shape constraints and credibility of identification. (a) Five-fold out-of-sample MSE: the shape-constrained NPIML matches the linear baselines' fit while the equal-capacity unconstrained MLP overfits; NPIML is 100% monotone. (b) Phase-0 kill switch: curve-relative price response falls with excess-supply shocks ($\beta = -0.028$, $t = -3.27$).

5.3 The impact surface

Figure 2 plots the estimated $\Lambda(Q, s)$ over the demand shock and the dealer-constraint axis $C = -$take-down. The surface passes through the origin, is monotone, and visibly steepens—both with the magnitude of the shock and as constraints tighten.

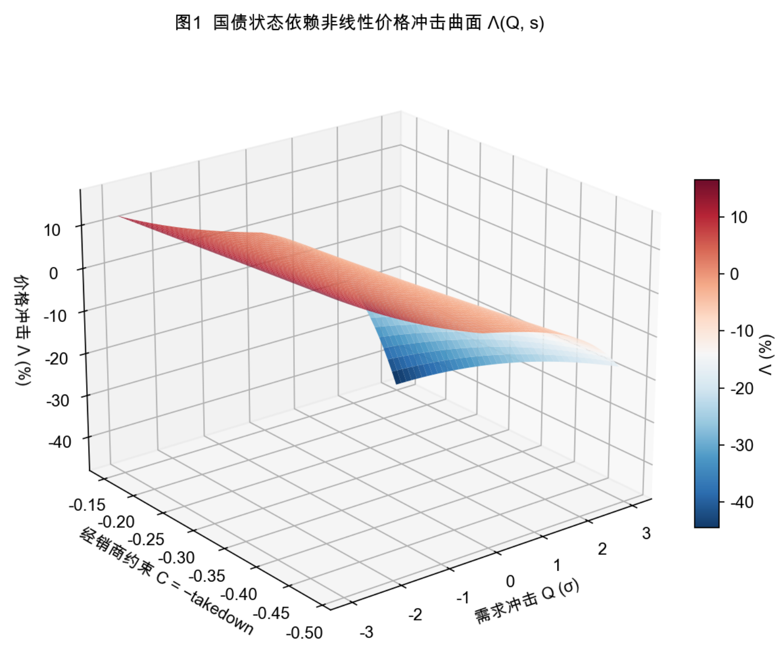

Figure 2: Estimated state-dependent nonlinear price impact surface $\Lambda(Q, s)$ for Treasuries. Price impact (%) as a function of the normalized demand shock Q (σ units) and the dealer-constraint state $C = -$take-down.

5.4 H1–H2: constraints reshape level *and* curvature

Quarterly regressions ($N = 67$, 2008Q2–2024Q4) of the state-implied measures on the dealer-constraint proxy C (auction take-down share, high = constrained) with credit and volatility controls give:

Hypothesis	Estimand	β_C	t
H1 (level)	λ	+0.0091	4.32
H2 (curvature)	κ	+0.0119	9.18
H3 (shape gap at 2σ)	gap	+0.0023	2.71

Both the level and the curvature rise with constraints, and the curvature effect is the stronger and more precisely estimated of the two. Independently, VIX loads positively on curvature ($t = 2.62$). Figure 3 shows the mechanism directly: under tight constraints the impact function is barely distinguishable from the loose-constraint case for small shocks but fans out sharply in the tails—exactly the object a linear measure cannot see.

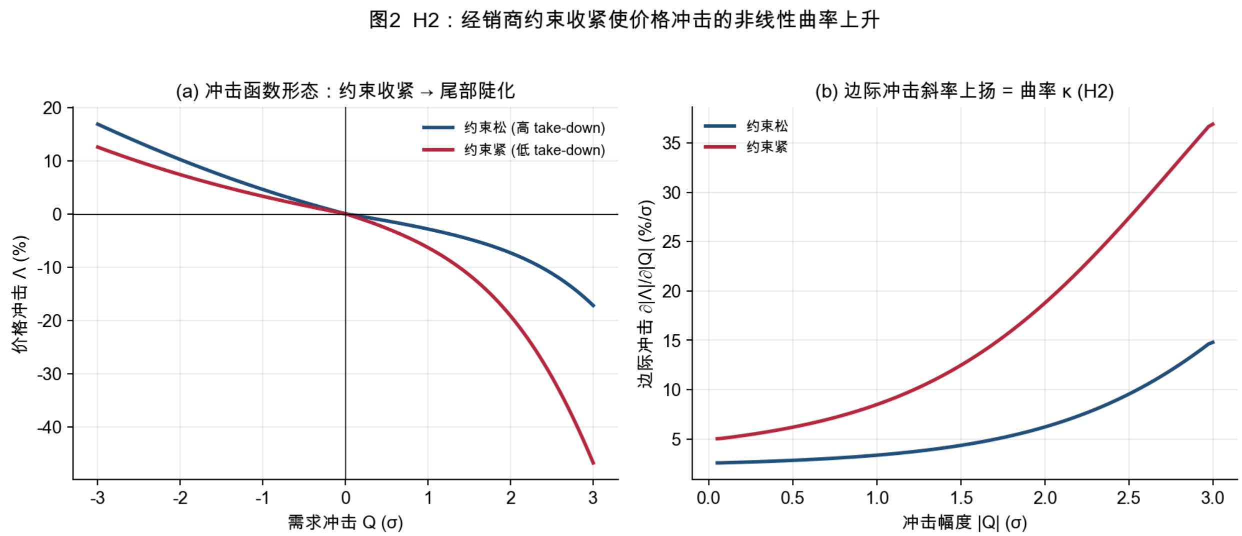

Figure 3: H2: tightening dealer constraints raise the nonlinear curvature of price impact. (a) The estimated impact function under loose vs. tight constraints; the curves nearly coincide near the origin and diverge in the tails. (b) The marginal-impact slope $\partial|\Lambda|/\partial|Q|$ rises faster under tight constraints—the curvature κ.

5.5 Basel III as a natural experiment

The post-2014 tightening of the Supplementary Leverage Ratio provides a clean break in dealer balance-sheet capacity. Mean curvature rises from 0.013 (pre-2014) to 0.030 (post-2014), $\Delta\kappa = +0.017$ ($t = 6.48$, $p < 10^{-10}$; Figure 4), consistent with H2 and not attributable to a contemporaneous change in the level alone.

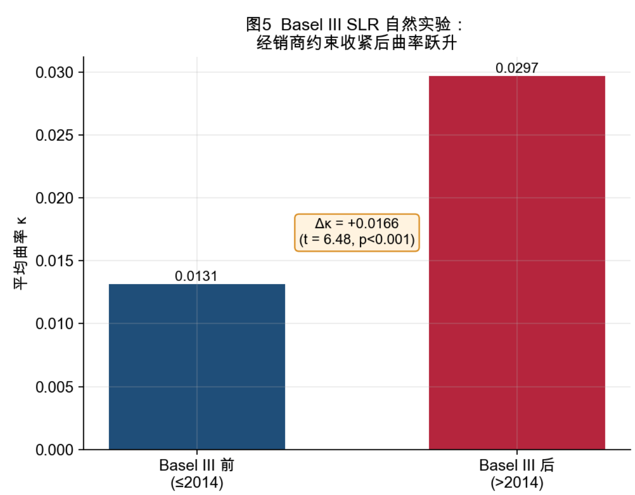

Figure 4: Basel III SLR natural experiment. Mean impact curvature κ before vs. after the 2014 leverage-ratio tightening.

5.6 Term structure and crisis precursors

Curvature is not uniform across the curve. Splitting into short (2–3Y), belly (5–7Y) and long (10–30Y) buckets, the stress-minus-normal curvature increment is essentially zero at the short end and belly but +0.075 at the long end (Figure 5): fragility is a long-duration phenomenon. In the time series, curvature behaves as a precursor. Around the March-2020 dislocation the standardized curvature κ_z was already elevated (+0.7 to +1.0) through 2019 while the level λ_z sat near zero; through 2022 κ_z ran near +2 (Figure 6). The nonlinear curvature carries fragility information that the linear level measure does not.

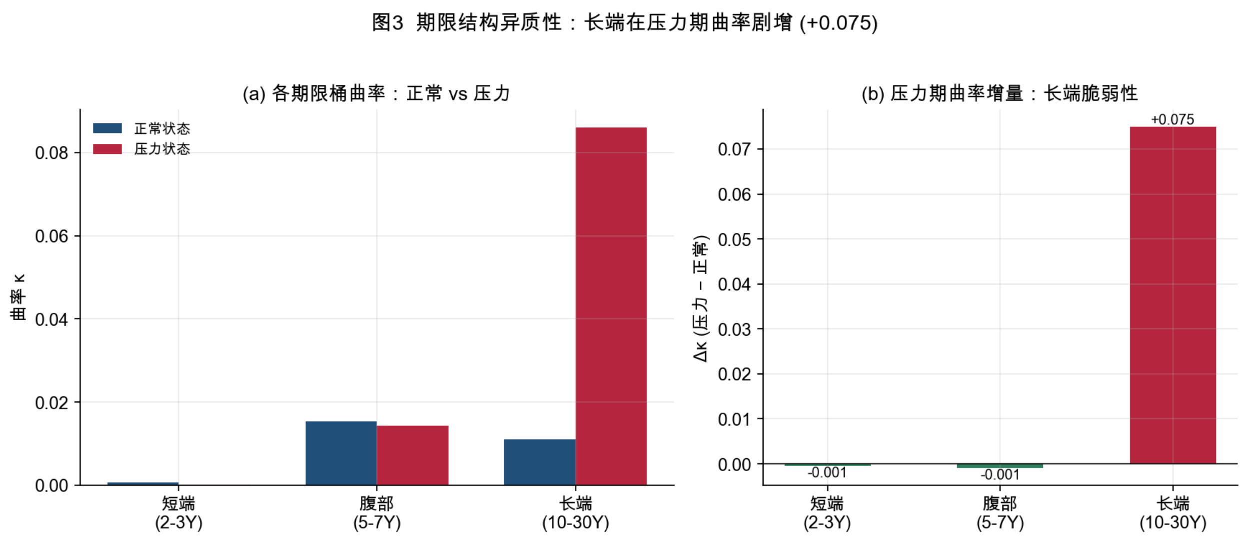

Figure 5: Term-structure heterogeneity. (a) Curvature κ by maturity bucket in normal vs. stress states. (b) Stress-minus-normal increment: the long end drives the fragility.

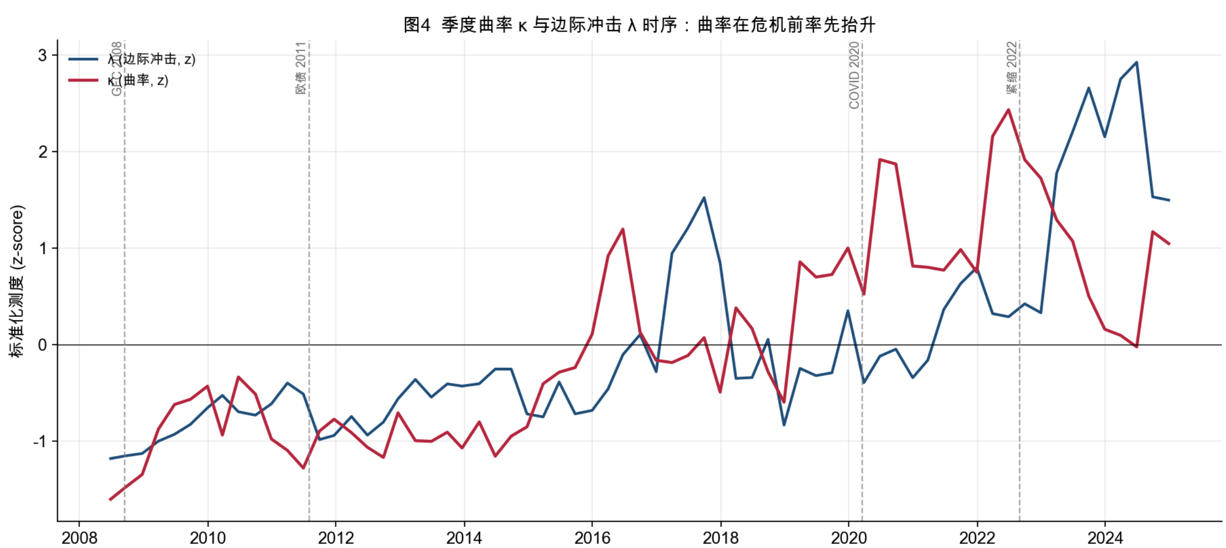

Figure 6: Curvature as a crisis precursor. Standardized quarterly curvature κ and marginal impact λ; dashed lines mark major liquidity events. Curvature rises ahead of realized crises while the level stays flat.

5.7 On-chain out-of-sample validation

Figure 7 reports the on-chain replication. Across 40,360 forced-liquidation events on eight perpetual markets the estimated Λ is 100% monotone; local slope crosses 1.2$\times$ its baseline at $Q^\star \approx 3.3\sigma$, a critical threshold beyond which forced-flow impact accelerates. This is a directly usable systemic-liquidity risk parameter: it flags the shock magnitude at which a venue's depth stops absorbing flow linearly.

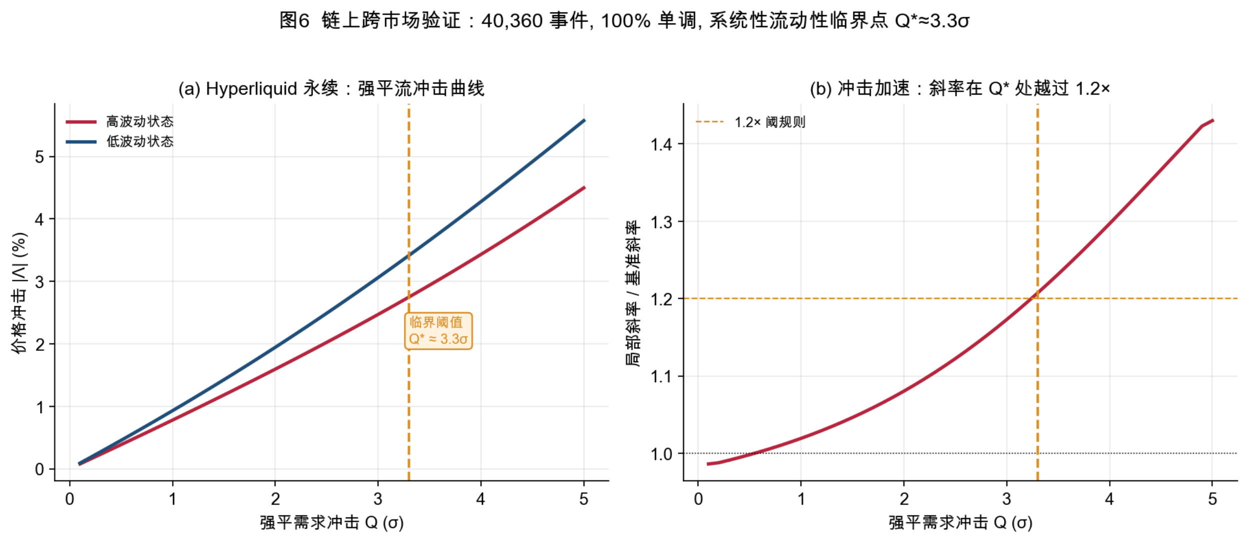

Figure 7: On-chain cross-market validation (Hyperliquid perpetuals). (a) Forced-liquidation impact curves in high- vs. low-volatility states. (b) Local slope relative to baseline crosses 1.2$\times$ at $Q^\star \approx 3.3\sigma$.

6 Robustness and Identification Threats

The exclusion restriction requires that the auction demand surprise be orthogonal to same-day private information about the tenor. Three features support it: (i) we difference out the common curve factor, so any economy-wide information that moves all yields is removed; (ii) the pure size-surprise instrument, which is set by the pre-announced financing calendar, carries the expected sign; and (iii) the tight $[-1, +1]$ window limits contamination from slow-moving information. The kill switch and the Basel natural experiment provide two independent confirmations that do not rely on the network. Remaining threats—auction-timing predictability and tenor-specific demand clienteles—are mitigated by the curve-relative differencing and the tenor controls but are not fully eliminated; we flag them as directions for instrumenting the demand surprise with primary-dealer positioning data.

7 Conclusion

Liquidity in markets without an order book can be measured by turning the problem around: instead of inferring liquidity from endogenous price movements, estimate the price response to exogenous demand shocks and treat that response function as the definition of liquidity. Doing so with shape-constrained learning makes the *curvature* of price impact—not just its slope—an interpretable estimand, and that curvature turns out to be where dealer constraints bite, where the long end is fragile, and where crises announce themselves in advance. The framework is portable: it transfers cleanly to on-chain markets where high-frequency ground truth confirms it. For markets from Treasuries to corporate bonds to swaps, a curvature-aware, state-dependent impact function offers a liquidity measure that is both economically interpretable and computable from public low-frequency data.

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